Sequences & Patterns

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

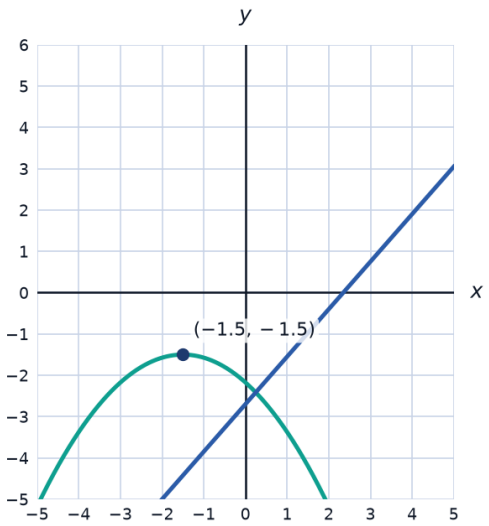
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Arithmetic sequences

Find the 10th term of the sequence $5, 8, 11, 14, \ldots$

2 [Foundational] • Target: Arithmetic sequences

An arithmetic sequence has first term 4 and common difference 6. Find the 20th term.



3 [Foundational] • Target: Arithmetic sequences

The 5th term of an arithmetic sequence is 23, and the first term is 3. Find the common difference.

4 [Foundational] • Target: Arithmetic sequences

An arithmetic sequence has first term 7 and common difference 5. Which term equals 82?

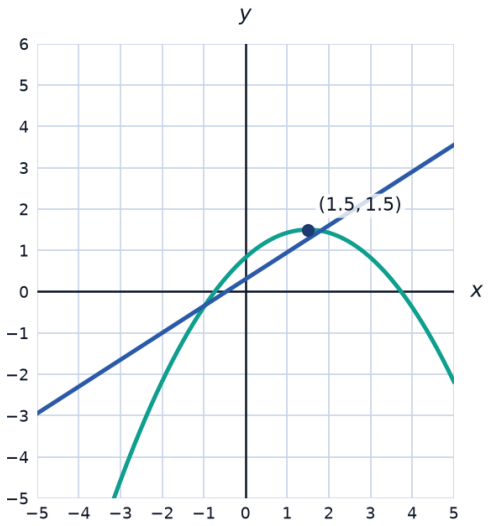
5 [Foundational] • Target: Arithmetic sequences

The 8th term of an arithmetic sequence is 34, and the common difference is 4. Find the first term.

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6 [Foundational] • Target: Geometric sequences

Find the next term of the sequence $2, 4, 8, 16, \ldots$



7 [Foundational] • Target: Geometric sequences

A geometric sequence has first term 5 and common ratio 3. Find the 5th term.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Geometric sequences

The 4th term of a geometric sequence is 54, and the first term is 2. Find the common ratio.

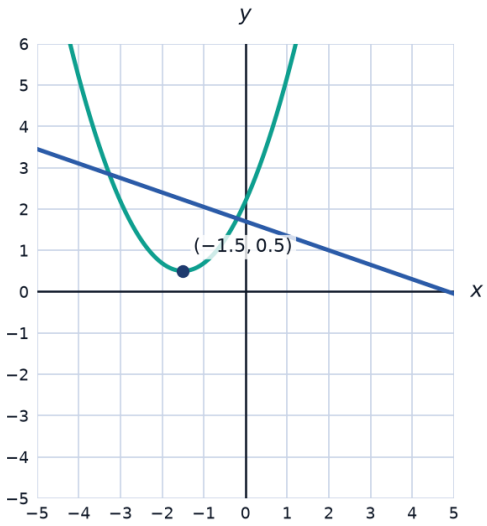
9 [Intermediate] • Target: Geometric sequences

A geometric sequence has 2nd term 6 and 3rd term 18. Find the first term and the common ratio.

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10 [Intermediate] • Target: Identifying the type of sequence

Is $5, 10, 20, 40, \ldots$ arithmetic, geometric, or neither?



11 [Intermediate] • Target: Finding missing terms

Find the missing term: $3,\ __,\ 11,\ 15$ (arithmetic).

12 [Intermediate] • Target: Finding missing terms

Find the missing term: $4,\ __,\ 36$ (geometric, all terms positive).

13 [Intermediate] • Target: Finding missing terms

Insert two arithmetic means between 5 and 20, forming the four-term sequence $5, __, __, 20$.

14 [Intermediate] • Target: Finding missing terms

The sequence $x,\ 2x + 1,\ 4x - 1$ is arithmetic. Find x.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

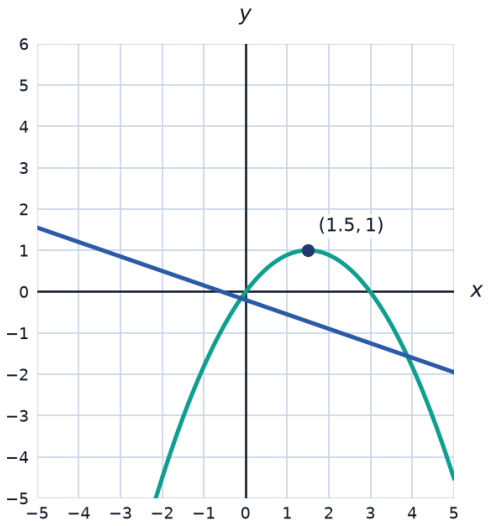
15 [Challenging] • Target: Finding missing terms • Cross-Lesson: 7.01 Data Analysis & Probability

Find the two missing terms of the geometric sequence $2,\ __,\ __,\ 54$.

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16 [Challenging] • Target: Sum of an arithmetic series • Cross-Lesson: 8.01 Word Problems & Reasoning

Find the sum of the first 10 terms of the sequence $2, 5, 8, 11, \ldots$



17 [Challenging] • Target: Sum of an arithmetic series • Cross-Lesson: 1.01 Arithmetic & Number Sense

Find the sum of the first 20 positive integers.

18 [Challenging] • Target: Sum of an arithmetic series • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

An arithmetic sequence has first term 3 and common difference 4. Find the sum of its first 15 terms.

19 [Challenging] • Target: Sum of an arithmetic series • Cross-Lesson: 3.01 Algebra Essentials

Find the sum of the first 25 positive even integers $(2, 4, \ldots, 50)$.

20 [Challenging] • Target: Special sequences • Cross-Lesson: 4.01 Geometry & Measurement

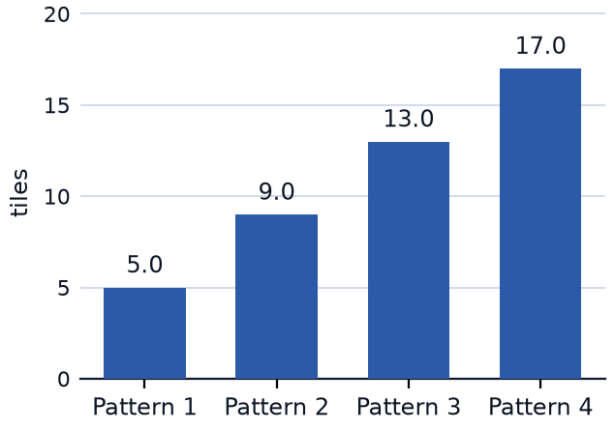
The sequence $1, 4, 9, 16, 25, \ldots$ consists of perfect squares. Find the 8th term.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Visual and growing patterns

The bar chart shows the number of tiles used to build patterns 1 through 4. Predict the number of tiles in pattern 8.



a. 37

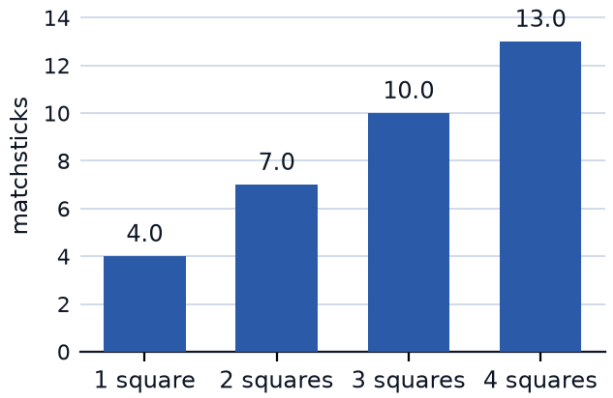
b. 40

c. 29

d. 33

22 [Foundational] • Target: Visual and growing patterns

The bar chart shows the number of matchsticks used to build squares in a row, for 1 through 4 squares. Find the number needed for 12 squares.



a. 34

b. 43

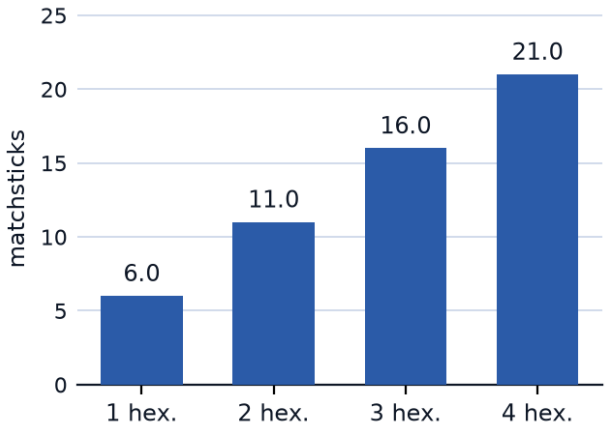
c. 40

d. 37

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23 [Foundational] • Target: Visual and growing patterns

The bar chart shows the number of matchsticks used to build hexagons in a row, for 1 through 4 hexagons. Find the number needed for 9 hexagons.



a. 41

b. 51

c. 50

d. 46

24 [Foundational] • Target: Geometric sequences

A geometric sequence has 3rd term 20 and 4th term 40. Find the first term and the common ratio.

a. $a_1 = 5, r = 2$

b. $a_1 = 20, r = 2$

c. $a_1 = 5, r = 4$

d. $a_1 = 10, r = 2$

25 [Foundational] • Target: Finding missing terms

The sequence $2x,\ 3x + 4,\ 5x + 2$ is arithmetic. Find x.

a. 4

b. 2

c. 8

d. 6

26 [Foundational] • Target: Finding missing terms

Find the missing term: __, 12, 19, 26 (arithmetic).

a. 7

b. 9

c. 4

d. 5

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27 [Foundational] • Target: Special sequences

Find the 7th triangular number, from the sequence $1, 3, 6, 10, 15, 21, \ldots$

a. 28

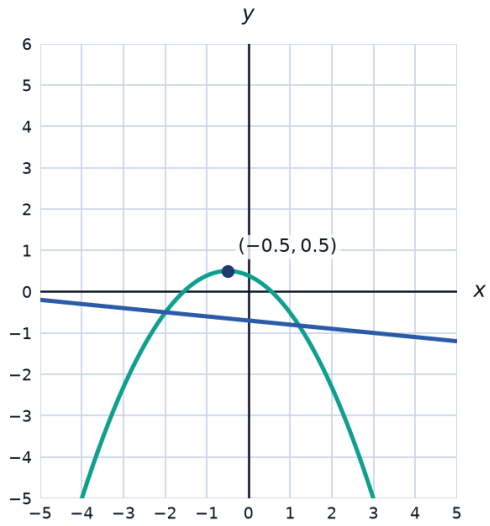
b. 25

c. 36

d. 21

28 [Foundational] • Target: Special sequences

A sequence starts $3, 5, 8, 13, 21, \ldots$, where each term is the sum of the two terms before it. Find the 8th term.



a. 55

b. 89

c. 144

d. 76

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Word problems

A theater has 18 seats in the first row, and each following row has 4 more seats than the row before. How many seats are in row 15?

a. 78

b. 70

c. 56

d. 74

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30 [Intermediate] • Target: Word problems

A ball is dropped and after each bounce rises to half its previous height. If it first rises to 120 cm after the first bounce, how high does it rise after the 5th bounce?

a. 60

b. 15

c. 3.75

d. 7.5

31 [Intermediate] • Target: Word problems

A savings plan deposits 80 riyals in week 1, then increases the deposit by 15 riyals each following week. Find the deposit in week 10.

a. 215

b. 200

c. 230

d. 935

32 [Intermediate] • Target: Word problems

A bacteria culture starts with 40 bacteria and triples every hour. How many bacteria are there after 4 hours?

a. 1,080

b. 3,240

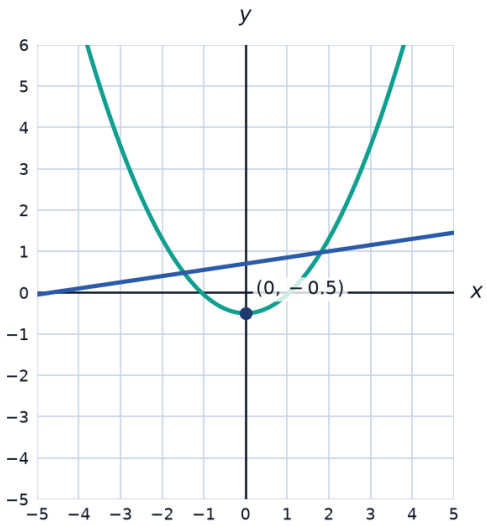
c. 9,720

d. 480

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33 [Intermediate] • Target: Pattern reasoning

The number of diagonals in a polygon with n sides is $\frac{n(n-3)}{2}$. Find the number of diagonals in a polygon with 9 sides.



a. 36

b. 27

c. 21

d. 54

34 [Intermediate] • Target: Pattern reasoning

A sequence follows the rule \"triple and subtract 2\", starting at 4 as the 1st term. Find the 4th term.

a. 34

b. 28

c. 82

d. 250

35 [Intermediate] • Target: Pattern reasoning

The nth term of a sequence is $T_n = 4n - 3$. Find the 15th term, and state whether the sequence is arithmetic or geometric.

a. 57, geometric

b. 53, arithmetic

c. 60, arithmetic

d. 57, arithmetic

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36 [Intermediate] • Target: Pattern reasoning

The nth term of a sequence is $T_n = 5(2)^{n-1}$. Find the 6th term, and state whether the sequence is arithmetic or geometric.

a. 160, geometric

b. 320, geometric

c. 160, arithmetic

d. 80, geometric

37 [Intermediate] • Target: Sequences & Patterns

In a laboratory calibration sequence, find the 13th term of the sequence $-14, -21, -28, -35, \ldots$

a. -98

b. -105

c. -96

d. -91

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Sequences & Patterns • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

During an engineering benchmark, find the 20th term of the sequence $1, -3, -7, -11, \ldots$

a. -71

b. -75

c. -73

d. -79

39 [Challenging] • Target: Sequences & Patterns • Cross-Lesson: 3.01 Algebra Essentials

In a telemetry monitoring trial, find the 7th term of the sequence $-5, -1, 3, 7, \ldots$

a. 23

b. 19

c. 21

d. 15

40 [Challenging] • Target: Sequences & Patterns • Cross-Lesson: 4.01 Geometry & Measurement

While auditing system throughput, find the 6th term of the sequence $-2, 1, 4, 7, \ldots$

a. 10

b. 13

c. 16

d. 15

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41 [Challenging] • Target: Sequences & Patterns • Cross-Lesson: 5.01 Quantitative Comparisons

Find the 3rd term of the sequence $-3, 6, -12, 24, \ldots$

a. 12

b. -10

c. -14

d. -12

42 [Challenging] • Target: Sequences & Patterns • Cross-Lesson: 7.01 Data Analysis & Probability

Under steady-state operating conditions, find the 7th term of the sequence $-1, 3, -9, 27, \ldots$

a. -729

b. 729

c. -727

d. -731

43 [Challenging] • Target: Sequences & Patterns • Cross-Lesson: 8.01 Word Problems & Reasoning

Across successive production cycles, find the 5th term of the sequence $3, -9, 27, -81, \ldots$

a. -243

b. 245

c. 241

d. 243

44 [Challenging] • Target: Sequences & Patterns • Cross-Lesson: 1.01 Arithmetic & Number Sense

In a precision metrology test, find the 3rd term of the sequence $7, 21, 63, 189, \ldots$

a. 61

b. 65

c. 63

d. -63

45 [Challenging] • Target: Sequences & Patterns • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

For a control-loop diagnostic run, find the 3rd term of the sequence $-1, -3, -9, -27, \ldots$

a. 9

b. -9

c. -7

d. -11

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